

Cover exactingness between strongly compact cardinals

A conditional inconsistency theorem from ω -club amenability and stationary seeds

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Abstract. We continue the completeness-barrier analysis in the supplied *Large Cardinals Synthesis*. The additional observation is that $\text{HCD}(\eta)$ contains the trace of every ambient ω -club filter. A stationary-seed argument then shows that an ω -club-amenable set-forcing ground must have the δ -cover property whenever δ is strongly compact in the extension. Such a ground cannot regard an ordinal above δ as regular if it has ambient cofinality ω . Combining this with the inherited completeness barrier and Goldberg’s ground theorem gives the following conditional inconsistency.

Main result: the two strongly compact cardinals cannot coexist with the cover

There are no cardinals

$$\delta < \lambda \leq \gamma < \kappa$$

such that δ and κ are strongly compact and λ is γ -cover exacting. Equivalently,

$$\text{ZFC} \vdash \neg \exists \delta, \lambda, \gamma, \kappa \left[\begin{array}{l} \delta < \lambda \leq \gamma < \kappa \wedge \text{SC}(\delta) \\ \wedge \text{CE}_{\gamma}(\lambda) \wedge \text{SC}(\kappa) \end{array} \right].$$

No Ground Axiom, stabilization of the HCD hierarchy, or preservation of a particular successor cardinal is assumed.

We also obtain a local strengthening. If $\delta < \lambda$ is strongly compact, $\text{CE}_{\gamma}(\lambda)$ holds, and a high-completeness model $N = \text{HCD}(\eta)$ satisfies Choice, then every sufficiently high ambient regular cardinal is measurable in N . More precisely, it suffices that $\theta > (2^{\lambda})^N$. The proof bounds the number of stationary cells that can survive in an N -partition and includes a proof of the classical filter-atomicity lemma used in this last step.

Status and attribution. These are new deductions *relative to the supplied synthesis*, not claims of historical priority. The stationary-seed method and the underlying HOD dichotomy are due to Goldberg. The cover-exacting characterizations are due to Blue–Goldberg. Detailed English proofs of the deductions are supplied; major external inputs are explicitly separated from them. This report is unrefereed and has not been formally verified in Lean. It does not refute cover exactingness by itself, nor does it settle the unrestricted problem of a cover-exacting cardinal above just one strongly compact cardinal.

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The argument in one line

$$\begin{aligned}
 \text{CEX}_\gamma(\lambda) &\implies \text{cf}^{\text{HCD}(\kappa)}(\lambda) = \lambda, & \text{cf}^V(\lambda) &= \omega, \\
 \text{SC}(\kappa), \kappa > \gamma &\implies \text{HCD}(\kappa) \text{ is a ground,} \\
 \text{SC}(\delta), \delta < \lambda, \omega\text{-club amenability of that ground} &\implies \text{cf}^{\text{HCD}(\kappa)}(\lambda) < \delta.
 \end{aligned}$$

The contradiction is between the first and last lines. The additional covering in the last line comes from stationary seeds; it is not an illicit increase in the completeness of a given ultrafilter.

1 What is added to the previous research

The supplied synthesis [1], especially its completeness barrier and its sections on a strongly compact cardinal above the cover bound, already proves

$$\text{CEx}_\gamma(\lambda) \implies \text{cf}^V(\lambda) = \omega \quad \text{and} \quad \text{cf}^{\text{HCD}(\eta)}(\lambda) = \lambda \quad (\eta \geq q(\gamma)), \quad (1.1)$$

where

$$q(\gamma) = \begin{cases} \gamma, & \gamma \text{ is singular,} \\ \gamma^+, & \gamma \text{ is regular.} \end{cases}$$

It also observes that an upper strongly compact $\kappa > \gamma$ makes $\text{HCD}(\kappa)$ a proper ground. With a lower strongly compact $\delta < \lambda$, its “two strongly compacts” corollary gives strict separation between two complete-definability models. Its later “collapse or destroy” theorem leaves a disjunction between collapsing the old successor and destroying a particular stationary set.

The missing additional ingredient is not a stronger completeness assumption. It is the fact that the high-completeness model can *recognize the trace of ambient ω -club positivity*. This permits one to put a small stationary seed *inside the image of that model*. A lower strongly compact cardinal then supplies small covers in the high model, contradicting (1.1).

Theorem 1.1 (No strongly compact sandwich). In ZFC, suppose $\delta < \lambda \leq \gamma$ are cardinals, δ is strongly compact, and $\text{CEx}_\gamma(\lambda)$ holds. Then no strongly compact cardinal is greater than γ .

The proof appears in Section 7, after a general no-ground theorem. A second, shorter verification uses Goldberg’s published covering dichotomy in Section 7.1.

How the earlier alternatives change

Under *both* strongly compact hypotheses, the forcing extension postulated in the earlier analysis does not exist. Thus we do not choose the collapse branch or the stationary-destruction branch of that analysis: we rule out their common premise. Without the upper strongly compact, the original coexistence question is not decided here.

A separate new consequence, which does not assume an upper strongly compact, is the quantitative stationary-width theorem of Section 8. Whenever a high HCD model satisfies ZFC, its ambient ω -club filters have no internally indexed disjoint positive family of length λ . This yields an explicit threshold for measurability in that inner model. The Axiom of Choice in the high model is an actual hypothesis in this consequence; it is not silently inferred from the lower strongly compact.

2 Definitions and the external inputs

2.1 Ambient and internal conventions

The ambient universe V satisfies ZFC. An inner model is transitive, contains all ordinals, and is a class of V . Unadorned cardinalities, ultrafilter completeness, cofinalities, and stationarity are ambient. A superscript N specifies computation in N . In particular, $P_\delta(\mu) = \{a \subseteq \mu : |a|^V < \delta\}$.

We use only set-parameter-definable inner models in arguments involving an ultrapower. If $N = \{x : \varphi(x, p)\}$ and $j : V \rightarrow M$ is an ultrapower embedding, the notation N^j means

$$N^j = \{x \in M : M \models \varphi(x, j(p))\}.$$

It does *not* mean the generally nontransitive pointwise image $j''N$. For $N = \text{HCD}^V(\eta)$, it is exactly $N^j = \text{HCD}^M(j(\eta))$. All uses of j below arise from a set ultrafilter; the relevant restrictions of j and images of sets are sets definable from that ultrafilter. No additional axiom asserting a class embedding is being imposed.

A ground is an inner model $W \models \text{ZFC}$ such that $V = W[G]$ for a set forcing $\mathbb{P} \in W$ and a W -generic filter G . A δ -cover of x in N is a set $a \in N$ with $x \subseteq a$ and $|a|^N < \delta$.

Lemma 2.1 (Comparing smallness). If $N \models \text{ZFC}$ is an inner model, δ is a cardinal of V , and $a \in N$, then $|a|^V < \delta$ if and only if $|a|^N < \delta$.

Proof. The ordinal δ is also a cardinal in N , since a bijection in N witnessing otherwise would remain a bijection in V . An N -bijection from an ordinal below δ onto a proves ambient smallness. Conversely, if N regarded a as having size at least δ , Choice in N would give an injection $\delta \rightarrow a$ in N , and hence in V . That contradicts $|a|^V < \delta$. $\square$

2.2 Cover exactingness and complete definability

For $\alpha > \lambda$ and $Y \subseteq V_\alpha$, a relative λ -exacting witness is an elementary embedding

$$j : (X, \in, X \cap Y) \longrightarrow (V_\alpha, \in, Y)$$

where the source is an elementary substructure of the target, $V_\lambda \cup \{\lambda\} \subseteq X$, $j(\lambda) = \lambda$, and $j \upharpoonright \lambda \neq \text{id}$. The domain need not be transitive. Write $\text{REx}(\lambda, Y)$ when such witnesses exist at all sufficiently large heights containing the predicate Y . The equivalent all-heights version may also be used.

Definition 2.2 (Uniform cover bound). For infinite cardinals $\lambda \leq \gamma$, $\text{CEX}_\gamma(\lambda)$ means that for every $\alpha > \lambda$ and every $y \in V_\alpha$, some $Y \subseteq V_\alpha$ satisfies $y \in Y$, $|Y| \leq \gamma$, and admits a relative λ -exacting witness at height α .

Here γ is fixed before the height and the parameter are chosen; $y \in Y$ must not be replaced by $y \subseteq Y$. Ordinary exactingness is obtained by forgetting the extra predicate and cover requirement.

For an infinite cardinal η , $\text{CD}(\eta)$ consists of sets definable in V from finitely many ordinal parameters and finitely many η -complete ultrafilters on ordinals. Completeness means closure under intersections of fewer than η members. Put

$$\text{HCD}(\eta) = \{x : \text{tc}(\{x\}) \subseteq \text{CD}(\eta)\}.$$

These definitions agree with Goldberg's complete-definability classes [3, §4.1]. We have

$$\eta \leq \xi \implies \text{CD}(\xi) \subseteq \text{CD}(\eta), \quad \text{HCD}(\xi) \subseteq \text{HCD}(\eta). \quad (2.1)$$

Every ordinal-definable set belongs to $\text{CD}(\eta)$. For a set of ordinals a , membership in $\text{CD}(\eta)$ is equivalent to membership in $\text{HCD}(\eta)$.

2.3 Imported theorems, with their exact roles

Input A (Local Kunen inconsistency). There is no nontrivial elementary embedding $V_{\rho+2} \rightarrow V_{\rho+2}$ in ZFC [7].

Input B (Blue–Goldberg). The following consequences of cover exactingness are used [2, Lemma 3.3 and Proposition 3.4]. If $\text{REx}(\lambda, Y)$ and $z \in \text{OD}_{\{Y\}}$, then $\text{REx}(\lambda, z)$. Also, if $\text{CE}_{X_Y}(\lambda)$ holds, then for all sufficiently large α the set

$$\{\sigma \in P_{Y^+}(V_\alpha) : \text{REx}(\lambda, \sigma)\}$$

is stationary. In particular it meets the club of elementary hulls containing any prescribed finite set of parameters.

These are explicitly stated lecture-note inputs, not a newly claimed consistency construction. The supplied synthesis also gives a persistent-cover proof of the exact-hull form of this input. We use the stationary form to keep the inherited portion short, and reprove the trace argument in Section 3.

Input C (Goldberg’s complete-definability ground theorem). For every infinite η , $\text{HCD}(\eta) \models \text{ZF}$. If κ is strongly compact, then $\text{HCD}(\kappa) \models \text{ZFC}$ and is a set-forcing ground of V [3, Proposition 4.2, Theorems 4.3 and 4.9, arXiv v2].

The theorem numbers in the journal version are shifted: the ground statement is Theorem 4.10. The arXiv v2 statement and proof were checked directly. Strong compactness, not supercompactness, suffices. The stronger hypothesis in the shorter 2026 lecture-note summary should not be substituted for the actual hypothesis of this theorem. No bound on the size of the ground forcing is used in the main result.

We also use the usual ultrafilter formulation of strong compactness: for every ordinal $\theta \geq \delta$, a strongly compact δ admits a δ -complete fine ultrafilter on $P_\delta(\theta)$. This is the form needed by the stationary-seed method in [4, Proposition 2.3]. Fineness means that $\{a : \xi \in a\}$ is in the ultrafilter for every $\xi < \theta$. Countable completeness makes its ultrapower well founded. These standard large-cardinal and ultrapower facts are external background; we do not present them as additional new results.

3 Reconstructing the inherited completeness barrier

This section identifies exactly what is carried over from the supplied reports. The rest of the paper needs only its final conclusion.

Lemma 3.1 (Fixed ordinals of a rank self-embedding). Let λ be an uncountable cardinal and let $e : V_\lambda \rightarrow V_\lambda$ be nontrivial. If $\kappa_0 = \text{crit}(e)$ and $\kappa_{n+1} = e(\kappa_n)$, then

$$\sup_{n < \omega} \kappa_n = \lambda, \quad e(\xi) > \xi \quad (\kappa_0 \leq \xi < \lambda).$$

Consequently $\text{cf}^V(\lambda) = \omega$, and every fixed ordinal below λ is below $\text{crit}(e)$.

Proof. Suppose $\rho = \sup_n \kappa_n < \lambda$. The critical sequence and the sets needed to define its supremum belong to V_λ : since λ is an uncountable cardinal, adding finitely many ranks, or ω ranks, to an ordinal below λ still stays below λ . Applying e to the sequence shifts it by one place, so $e(\rho) = \rho$. The restriction of e to $V_{\rho+2}$ is elementary and nontrivial, contradicting Input A. Thus the critical sequence is cofinal in λ . If $\kappa_n \leq \xi < \kappa_{n+1}$, monotonicity on ordinals gives $e(\xi) \geq e(\kappa_n) = \kappa_{n+1} > \xi$. This proves both assertions. $\square$

An exacting witness restricts to such an e . Indeed V_λ is definable from the fixed parameter λ , is contained in its domain, and satisfaction for that rank segment is preserved. This argument does not assume that the full elementary submodel is transitive.

Lemma 3.2 (A short cofinal set cannot be a preserved predicate). If $a \subseteq \lambda$ is cofinal and $\text{ot}(a) < \lambda$, then $\text{REx}(\lambda, a)$ fails.

Proof. At a sufficiently large witnessing height, the set a is the unique set represented by its unary predicate. Thus $a \in X$ and $j(a) = a$. Its order type τ and its increasing enumeration $h : \tau \rightarrow a$ are definable from a , belong to X , and are fixed by j . Since $\tau < \lambda$ is fixed, Lemma 3.1 implies $\tau < \text{crit}(j)$. Every $\xi < \tau$ is therefore fixed, and

$$j(h(\xi)) = j(h)(j(\xi)) = h(\xi).$$

The cofinal set a would consist of fixed ordinals below λ , although all those fixed ordinals lie below $\text{crit}(j)$. This is impossible. $\square$

Lemma 3.3 (Reconstruction from a small trace). Suppose U is an η -complete ultrafilter on an ordinal τ , $U, \tau \in \sigma < V_\alpha$, the height is sufficiently large, and $|\sigma| < \eta$. Then $U \in \text{OD}_{\{\sigma\}}$.

Proof. The family $U \cap \sigma$ has size below η . Its intersection belongs to the proper ultrafilter U , so is nonempty. Let

$$b = \min \bigcap (U \cap \sigma).$$

For every $A \in \sigma$ with $A \subseteq \tau$, elementarity puts $\tau \setminus A$ in σ . The ultrafilter alternatives give

$$A \in U \iff b \in A. \quad (3.1)$$

Now U is the unique ultrafilter $W \in \sigma$ on τ with this trace. If two distinct candidates in σ had the same trace, elementarity would supply a set $A \in \sigma$ on which they differ, contradicting (3.1). The resulting definition uses only σ and the ordinals b, τ (and, if satisfaction is used, α). It therefore makes U ordinal definable from σ . Neither $b \in \sigma$ nor principality of U is required; uniqueness is only among the candidates inside σ . $\square$

Proposition 3.4 (Inherited barrier, with proof). Assume $\text{CEx}_\gamma(\lambda)$. For every cardinal $\eta \geq q(\gamma)$,

$$\text{cf}^V(\lambda) = \omega, \quad \text{cf}^{\text{HCD}(\eta)}(\lambda) = \lambda.$$

Moreover no short cofinal subset of λ belongs to $\text{CD}(\eta)$.

Proof. First let $\eta > \gamma$ and let $x \in \text{CD}(\eta)$, witnessed by finitely many η -complete ultrafilters $U_0, \dots, U_{m-1}$ on ordinals. Input **B** supplies an elementary hull $\sigma < V_\alpha$ containing these ultrafilters and their underlying ordinals, with $|\sigma| \leq \gamma$ and $\text{REx}(\lambda, \sigma)$. This follows by intersecting the stationary set there with the club of elementary hulls containing the listed parameters.

For each i , Lemma 3.3 applies because $|\sigma| \leq \gamma < \eta$. Substitution in the definition of x now gives $x \in \text{OD}_{\{\sigma\}}$. Definability transfer in Input **B** gives $\text{REx}(\lambda, x)$. Lemma 3.2 therefore excludes every cofinal $x \subseteq \lambda$ of order type below λ .

Put $N = \text{HCD}(\eta)$. The cardinal λ is still a cardinal in N . If N contained a cofinal map $f : \mu \rightarrow \lambda$ with $\mu < \lambda$, its range $a \in N \subseteq \text{CD}(\eta)$ would have ambient size below λ . As a subset of the initial ordinal λ , it would have order type below λ , a contradiction. This reasoning needs only ZF in N . Ambient countable cofinality follows from Lemma 3.1.

Finally suppose γ is singular. Every γ -complete filter is γ^+ -complete: a family of size γ can be partitioned into $\text{cf}(\gamma)$ many subfamilies of size below γ ; intersect within each subfamily and then intersect the resulting $\text{cf}(\gamma) < \gamma$ members. Thus $\text{CD}(\gamma) = \text{CD}(\gamma^+)$ and $\text{HCD}(\gamma) = \text{HCD}(\gamma^+)$. This includes the endpoint $\eta = q(\gamma) = \gamma$; monotonicity gives all larger η . $\square$

4 The missing amenability observation

For an ordinal α of uncountable cofinality define

$$E_\omega^\alpha = \{\xi < \alpha : \text{cf}^V(\xi) = \omega\}, \quad \mathcal{F}_{\alpha,\omega}^V = \{A \subseteq \alpha : \exists C \text{ club in } \alpha (C \cap E_\omega^\alpha \subseteq A)\}.$$

This is the filter generated by the ambient club filter and E_ω^α . A set $A \subseteq \alpha$ is positive for this filter precisely when $A \cap E_\omega^\alpha$ is stationary in α . The definition makes sense for singular α as long as $\text{cf}^V(\alpha) > \omega$.

Definition 4.1 (ω -club amenability). An inner model $N \models \text{ZFC}$ is ω -club amenable in V if, for every ordinal α with $\text{cf}^V(\alpha) > \omega$,

$$\mathcal{F}_{\alpha,\omega}^V \cap N \in N.$$

This is Goldberg's definition [4, §2.2]. The filter in this formula is the *ambient* filter, not N 's internally computed club filter.

The trace expression means the set of members of the filter that belong to N ; its members are subsets of α . The following observation applies even before Choice in N is available.

Lemma 4.2 (Ordinal-definable subsets of a hereditary class). If $N = \text{HCD}(\eta)$ and A is an ordinal-definable set with $A \subseteq N$, then $A \in N$.

Proof. The set A belongs to $\text{CD}(\eta)$ because every ordinal-definable set does. For each $x \in A$, the assumption $x \in N$ says that every member of $\text{tc}(\{x\})$ belongs to $\text{CD}(\eta)$. Therefore every member of $\text{tc}(\{A\})$ belongs to $\text{CD}(\eta)$, including A itself. This is exactly the condition $A \in \text{HCD}(\eta)$. $\square$

Proposition 4.3 (Amenability of the high-completeness models). For every infinite cardinal η and every ordinal α of uncountable ambient cofinality,

$$\mathcal{F}_{\alpha,\omega}^V \cap \text{HCD}(\eta) \in \text{HCD}(\eta).$$

In particular, whenever $\text{HCD}(\eta)$ satisfies Choice, it is an ω -club-amenable inner model of ZFC.

Proof. The trace is a set by Separation from $\mathcal{P}(\alpha)$. It is definable in V from the two ordinal parameters α, η , and all its members belong to $\text{HCD}(\eta)$. Lemma 4.2 applies. $\square$

Remark 4.4 (What this does not say). The ambient club or stationary sets themselves need not belong to N . Nor does this identify ambient stationarity with internal stationarity. It says that *for sets already in* N , the test of ambient ω -club positivity can be encoded by a set of N . This is exactly the amount of information required by the seed construction below.

5 Stationary lifting and the local seed lemma

The method is an explicit version of the stationary-seed arguments in Goldberg's work [4, Proposition 2.3 and Lemma 2.7]; compare also [3, Theorem 4.13] and [5, Proposition 2.2]. In contrast to the same-parameter HCD covering argument, the lower compactness bound and the higher complete-definability parameter are separated here. The essential additional bookkeeping is that the seed belongs to N^j , not merely to the ultrapower M . We prove the local form needed here in detail.

5.1 The continuity calculation

Lemma 5.1 (Lifting a stationary set of countable cofinalities). Let θ be an uncountable regular cardinal of V and $j : V \rightarrow M$ a well-founded ultrapower embedding with $j(\omega) = \omega$. Set $\beta = \sup j''\theta$. If $S \subseteq E_\omega^\theta$ is stationary in V , then $j(S) \cap \beta$ meets every club in β that belongs to M .

Proof. Fix such a club C . Closedness and unboundedness of C in the ordinal β are absolute to V . Define, externally in V ,

$$f(\xi) = \sup_{\zeta < \xi} j(\zeta) \quad (\xi < \theta).$$

At limit indices this function is continuous. Let

$$D = \{\xi < \theta : \xi \text{ is a nonzero limit ordinal and } f(\xi) \in C\}.$$

We verify directly that D is club in θ .

For unboundedness, start above any chosen $\xi_0 < \theta$ and recursively choose increasing $\xi_n < \theta$ and $c_n \in C$ so that

$$j(\xi_n) < c_n < j(\xi_{n+1}).$$

The choices are possible because both C and $j''\theta$ are unbounded in β . Regularity of θ gives $\xi = \sup_n \xi_n < \theta$. The two interleaved sequences have the same supremum, namely $f(\xi) = \sup_n c_n < \beta$; closedness of C puts $f(\xi)$ in C . Thus $\xi \in D$ and $\xi > \xi_0$. For closedness, if $D \cap \xi$ is unbounded in a limit $\xi < \theta$, continuity gives $f(\xi) = \sup\{f(\zeta) : \zeta \in D \cap \xi\} \in C$. The values here are unbounded in $f(\xi)$, so closedness applies.

Choose $\xi \in S \cap D$. A countable increasing cofinal sequence in ξ is carried by j to a countable cofinal sequence in $j(\xi)$, because $j(\omega) = \omega$. Consequently $j(\xi) = \sup j''\xi = f(\xi) \in C$. Also $j(\xi) \in j(S)$ and $j(\xi) < \beta$. This proves the assertion. $\square$

5.2 Constructing a seed inside the inner model

Theorem 5.2 (Local stationary-seed covering). Let $N \models \text{ZFC}$ be a set-parameter-definable, ω -club-amenable inner model. Suppose δ is strongly compact in V , μ is an ordinal, and $\theta > \max\{\delta, \mu\}$ is regular in V . Assume there is a sequence

$$\vec{A} = \langle A_\xi : \xi < \mu \rangle \in N$$

of pairwise disjoint subsets of θ , each positive for $\mathcal{F}_{\theta, \omega}^V$. Then there is a δ -complete fine ultrafilter U on $P_\delta(\mu)$ such that

$$N \cap P_\delta(\mu) \in U. \quad (5.1)$$

In particular every $x \subseteq \mu$ of ambient size below δ has a δ -cover in N .

Proof. Choose a δ -complete fine ultrafilter U_0 on $P_\delta(\theta)$ and form its well-founded ultrapower $j : V \rightarrow M$. Let $u = [\text{id}]_{U_0}$ be the canonical seed. Then

$$j''\theta \subseteq u \subseteq j(\theta), \quad |u|^M < j(\delta), \quad \text{crit}(j) \geq \delta. \quad (5.2)$$

Put $\beta = \sup j''\theta$. Since $j(\theta)$ is regular in M and u is M -small, $\beta < j(\theta)$.

Step 1: the cofinality bounds. The increasing sequence $j''\theta$ shows $\text{cf}^V(\beta) = \theta$. To see equality rather than just an upper bound, any cofinal subset of β of size below the regular cardinal θ could be bounded

by fewer than θ many corresponding points of $j''\theta$, contradicting cofinality of that sequence. Hence $\text{cf}^M(\beta) \geq \theta > \omega$. On the other hand $u \cap \beta \in M$ is cofinal in β and has M -size below $j(\delta)$. Thus

$$\omega < \text{cf}^M(\beta) < j(\delta). \quad (5.3)$$

Fix in M a club $C \subseteq \beta$ of size $\text{cf}^M(\beta) < j(\delta)$.

Step 2: the seed is definable in N^j . Let $\vec{B} = j(\vec{A}) = \langle B_\xi : \xi < j(\mu) \rangle$. It belongs to N^j and its entries are pairwise disjoint. By the transferred amenability statement and (5.3),

$$F = \mathcal{F}_{\beta, \omega}^M \cap N^j \in N^j.$$

For each $\xi < j(\mu)$, both $B_\xi \cap \beta$ and its complement in β belong to N^j . Therefore Separation *inside* N^j constructs the set

$$s = \{\xi < j(\mu) : \beta \setminus (B_\xi \cap \beta) \notin F\} \in N^j. \quad (5.4)$$

Equivalently, $\xi \in s$ exactly when $B_\xi \cap \beta$ is positive for the ambient-to- M filter $\mathcal{F}_{\beta, \omega}^M$. This is the point where the trace belongs to the smaller model; mere external definability of s in M would not suffice.

Step 3: fineness and smallness of the seed. Fix $\xi < \mu$. Since A_ξ is positive, $S = A_\xi \cap E_\omega^\theta$ is stationary in V . Lemma 5.1 shows that $j(S) \cap \beta$ meets every club of β in M . Every member of this set has M -cofinality ω , since $j(S) \subseteq (E_\omega^{j(\theta)})^M$. It is contained in $B_{j(\xi)} \cap \beta$. Therefore $j(\xi) \in s$, and we have

$$j''\mu \subseteq s. \quad (5.5)$$

Every $\xi \in s$ satisfies $B_\xi \cap C \neq \emptyset$. Choosing its least member defines in M an injection

$$\xi \mapsto \min(B_\xi \cap C) \quad \text{from } s \text{ into } C,$$

because the B_ξ are disjoint. Hence $|s|^M < j(\delta)$. Together with $s \subseteq j(\mu)$, this says $s \in j(P_\delta(\mu))$.

Step 4: derive the ultrafilter. Define in V

$$U = \{Z \subseteq P_\delta(\mu) : s \in j(Z)\}.$$

Since $s \in j(P_\delta(\mu))$, elementarity proves the ultrafilter axioms. The critical point bound in (5.2) proves δ -completeness: for a sequence of fewer than δ many members, j fixes the index ordinal and the seed lies in every entry of its image, hence in the image of the intersection. Equation (5.5) proves fineness. Finally the definition of N^j gives

$$j(N \cap P_\delta(\mu)) = N^j \cap j(P_\delta(\mu)).$$

Our seed belongs to the right-hand side by (5.4), proving (5.1).

Step 5: extract a cover. For $x \subseteq \mu$ with $|x| < \delta$, intersect the concentrating set (5.1) with the fine cones for all $\xi \in x$. There are fewer than δ cones, so the intersection is in the proper ultrafilter U and is nonempty. A member a belongs to N , contains x , and has ambient size below δ . Lemma 2.1 converts this to $|a|^N < \delta$. $\square$

Remark 5.3 (No unnecessary closure of the ultrapower). The proof does not assume $M^\theta \subseteq M$ or $j''\theta \in M$. The small set $u \in M$ covers that image; β is an ordinal and hence belongs to M . The lower bound on its cofinality is computed externally, while the upper bound is witnessed by $u \cap \beta \in M$. This is why strong compactness suffices where supercompactness might otherwise be mistakenly assumed.

6 Amenable grounds cannot regularize these ordinals

6.1 Why a ground supplies sufficiently high stationary families

Lemma 6.1 (High stationary partitions in a ground). Suppose $V = W[G]$ for a set forcing $\mathbb{P} \in W$, and μ is any ordinal. There is an ordinal $\theta > \mu$ that is a successor cardinal in W and regular in V , together with a sequence $\langle S_\xi : \xi < \mu \rangle \in W$ of pairwise disjoint subsets of $(E_\omega^\theta)^W$, each stationary in V . The ordinal θ can be chosen above any additional prescribed bound.

Proof. In W , choose an infinite cardinal ν greater than μ and $|\mathbb{P}|^W$, and beyond the prescribed bound. Put $\theta = (\nu^+)^W$. Then W regards $\mathbb{P}$ as θ -cc. Such forcing preserves regularity of θ and all W -stationary subsets of θ ; a direct proof is given in Appendix A.2.

Inside W , the set $(E_\omega^\theta)^W$ is stationary. The Ulam-matrix argument at a successor cardinal, proved in Appendix A.1, splits it into θ many disjoint stationary subsets. Take the first μ cells of the resulting sequence. They remain stationary in V . A countable cofinal sequence witnessing membership in $(E_\omega^\theta)^W$ remains a countable cofinal sequence in V , so all these cells are subsets of the actual E_ω^θ . $\square$

Theorem 6.2 (Covering by an ω -club-amenable ground). Let $W \models \text{ZFC}$ be a set-parameter-definable ground of V that is ω -club amenable in V . If δ is strongly compact in V , then W has the δ -cover property in V .

Proof. First let $x \subseteq \mu$ have size below δ . Apply Lemma 6.1, choosing the regular θ also above δ . The resulting sequence is a family of positive sets for $\mathcal{F}_{\theta, \omega}^V$, so Theorem 5.2 gives a cover $a \in W \cap P_\delta(\mu)$ of x . Lemma 2.1 gives $|a|^W < \delta$.

For the usual cover property on arbitrary sets $x \subseteq W$, choose an ordinal α with $x \subseteq V_\alpha^W$. Choice in W provides a bijection $b : \mu \rightarrow V_\alpha^W$ for some ordinal μ . The inverse image $b^{-1}[x]$ is an ambient set of ordinals of size below δ . Cover it by $a \in W$ of W -size below δ as just proved. Then $b[a] \in W$ covers x and still has W -size below δ . $\square$

Corollary 6.3 (No amenable-ground regularization). Suppose δ is strongly compact in V , $\lambda > \delta$ is an ordinal with $\text{cf}^V(\lambda) = \omega$, and W is a set-parameter-definable, ω -club-amenable inner model of ZFC in which λ is regular. Then W is not a set-forcing ground of V .

Proof. If it were a ground, take a countable cofinal $c \subseteq \lambda$ in V . Theorem 6.2 would give $a \in W$ containing c with $|a|^W < \delta$. Replace a by $a \cap \lambda$. This set is cofinal in λ because it contains c . Cofinality of a fixed set of ordinals is absolute between transitive models with the same ordinals. Thus W contains a cofinal subset of its regular cardinal λ of W -size below $\delta < \lambda$, a contradiction. $\square$

This argument is independent of the size, distributivity, or chain condition of the forcing at λ itself. Every set forcing has a high enough level at which it preserves the stationary partition needed by the proof. This is what makes the result stronger than a bound on a singularizing forcing at one specified successor.

7 The conditional inconsistency theorem

Theorem 7.1 (No high-completeness ground). Assume $\delta < \lambda \leq \gamma$, $\text{SC}(\delta)$, and $\text{CEX}_\gamma(\lambda)$. Then for every cardinal $\eta \geq q(\gamma)$, $\text{HCD}(\eta)$ is not a set-forcing ground of V .

Proof. Suppose $N = \text{HCD}(\eta)$ were a ground. In particular it would satisfy ZFC, whether or not Choice was known beforehand. Proposition 4.3 makes it ω -club amenable, and it is definable from the ordinal η . By Proposition 3.4, λ is regular in N while $\text{cf}^V(\lambda) = \omega$. Since $\delta < \lambda$ is strongly compact in V , Corollary 6.3 gives the contradiction. $\square$

Proof of Theorem 1.1. Suppose, contrary to the conclusion, that a strongly compact $\kappa > \gamma$ exists. Then $\kappa \geq q(\gamma)$. Input **C** says that $\text{HCD}(\kappa)$ is a set-forcing ground of V . Theorem 7.1 says that it is not. $\square$

Corollary 7.2 (A proper class of strongly compact cardinals). Assume there is a proper class of strongly compact cardinals, and let δ_0 be the least one. Every cover-exacting cardinal is strictly below δ_0 . Equivalently, no cardinal at or above δ_0 is γ -cover exacting for any cardinal γ .

Proof. If $\lambda > \delta_0$ were γ -cover exacting, choose a strongly compact $\kappa > \gamma$ and apply Theorem 1.1 with $\delta = \delta_0$. Equality $\lambda = \delta_0$ is impossible because an exacting cardinal has cofinality ω , whereas a strongly compact cardinal is uncountable regular. $\square$

Corollary 7.3 (A bound forced on every cover width). If a cover-exacting λ lies above a strongly compact cardinal, then the class of strongly compact cardinals is bounded. Every witness γ to $\text{CEx}_\gamma(\lambda)$ is an upper bound for that class. In particular, a λ -cover-exacting λ cannot have strongly compact cardinals both below and above it.

Proof. For each fixed cover bound γ , Theorem 1.1 excludes all strongly compact cardinals greater than γ . The diagonal case sets $\gamma = \lambda$. $\square$

Remark 7.4 (The strict inequality at the upper endpoint). The conclusion excludes $\kappa > \gamma$. It does not, by this argument, exclude $\kappa = \gamma$ when γ is regular. At a regular cover bound the inherited reconstruction requires completeness strictly greater than γ . The singular endpoint has already been included through $q(\gamma)$, but a strongly compact cardinal is not singular.

7.1 Cross-check against the published dichotomy

Goldberg proves the following result [4, Lemma 2.7]: if N is ω -club amenable and δ is ω_1 -strongly compact, then either all sufficiently large regular cardinals of V are measurable in N , or N has the (ω_1, δ) -cover property. The latter means that every countable subset of N is covered by a set in N of N -size below δ . A strongly compact cardinal satisfies the stated weaker hypothesis.

Here is a short second verification of the no-ground step. Suppose $V = N[G]$ by a set forcing. At arbitrarily large successor cardinals of N , the forcing preserves both that successor's cardinality and its regularity. In particular, there are arbitrarily large regular cardinals of V that are still successor cardinals in N . They cannot be measurable in N , since measurable cardinals are inaccessible in any model of ZFC. Thus the first alternative is impossible. The countable-cover alternative follows and contradicts regularity in N of a $\lambda > \delta$ of ambient cofinality ω , exactly as in Corollary 6.3.

This verification has the same classical stationary-seed background as our detailed proof; it is not a claim of a new HOD dichotomy. It is useful because it checks the main conclusion against an independently stated published theorem, and uses only countable covering rather than the full δ -cover property we proved for strongly compact δ .

8 Quantitative stationary width and inner measurability

The preceding proof has a local consequence even when no ground is assumed. It records how extensively high stationary partitions must fail in a hypothetical remaining model.

Theorem 8.1 (A bound on ambient stationary width). Let $N \models \text{ZFC}$ be set-parameter definable and ω -club amenable. Suppose δ is strongly compact in V , $\delta < \lambda$, $\text{cf}^V(\lambda) = \omega$, and λ is regular in N . For every regular cardinal $\theta > \lambda$ of V , the filter

$$F_\theta = \mathcal{F}_{\theta, \omega}^V \cap N \in N$$

is λ -weakly saturated in N : there is no sequence in N of λ pairwise disjoint F_θ -positive subsets of θ .

Proof. Such a sequence would meet all the assumptions of Theorem 5.2 with $\mu = \lambda$. A countable cofinal $c \subseteq \lambda$ would have an N -cover of N -size below δ , contradicting regularity of λ in N . For a set $A \in N$ with $A \subseteq \theta$, positivity for the filter object F_θ as computed in N is exactly ambient $\mathcal{F}_{\theta, \omega}^V$ -positivity: in both cases it says that $\theta \setminus A$ is not in the trace. Thus no absoluteness of stationarity has been assumed in translating the contradiction into the internal weak-saturation statement. $\square$

Corollary 8.2 (How many cells can survive?). Under the hypotheses of Theorem 8.1, let $\vec{S} = \langle S_\xi : \xi < \tau \rangle \in N$ be a sequence of pairwise disjoint subsets of θ , where $\theta > \lambda$ is regular in V . Then

$$I = \{\xi < \tau : S_\xi \text{ is positive for } \mathcal{F}_{\theta, \omega}^V\}$$

belongs to N and has N -cardinality below λ . In particular, if $\vec{S}$ is an N -partition of $(E_\omega^\theta)^N$ into θ stationary cells, fewer than λ of those cells remain stationary in V .

Proof. The trace $F_\theta \in N$ and the sequence $\vec{S} \in N$ define I by Separation in N , using the complement test for positivity. If $|I|^N \geq \lambda$, Choice in N would select a subsequence of length λ , contradicting Theorem 8.1. For the last assertion, every cell is a subset of the true E_ω^θ , so positivity is precisely ambient stationarity. $\square$

For $N = \text{HCD}(\eta)$ and a cover-exacting λ , the cardinal λ is also an ambient cardinal. In that application Lemma 2.1 converts the bound to ambient size below λ as well. Thus a partition of length $\theta > \lambda$ has θ many cells that cease to be stationary. This is a restriction at every applicable high level, not just at $(\lambda^+)^N$.

Lemma 8.3 (Internal completeness of the trace). If N is ω -club amenable and θ is uncountable regular in V , then N regards F_θ as a proper θ -complete filter containing every co-bounded subset of θ that belongs to N .

Proof. The ordinal θ remains a regular cardinal in N : any smaller cardinality or cofinality witness in N would remain such a witness in V . The ambient filter is proper because every club in a regular uncountable θ meets E_ω^θ . It contains the co-bounded sets.

Let $\langle A_i : i < \rho \rangle \in N$ with $\rho < \theta$ and each $A_i \in F_\theta$. In V choose a club C_i witnessing membership of each A_i in the ambient filter. Regularity of θ makes $C = \bigcap_{i < \rho} C_i$ a club. Then $C \cap E_\omega^\theta \subseteq \bigcap_{i < \rho} A_i$. This intersection belongs to N and to the ambient filter, hence to F_θ . All the filter axioms and the asserted internal completeness follow. The witness clubs need not belong to N . $\square$

Theorem 8.4 (An explicit inner-measurability threshold). Under the hypotheses of Theorem 8.1, every regular cardinal θ of V such that

$$\theta > (2^\lambda)^N$$

is measurable in N .

Proof. Work inside N . The ordinal λ is an infinite cardinal, F_θ is λ -weakly saturated by Theorem 8.1, and it is θ -complete by Lemma 8.3. Since $\theta > (2^\lambda)^N$, this supplies the $((2^\lambda)^+)^N$ -completeness needed by Ulam's atomicity lemma. The lemma, proved in Appendix B, partitions θ modulo the dual ideal into fewer than λ positive atoms. Restricting the filter to any one of these atoms yields a θ -complete ultrafilter in N on θ . It is nonprincipal: the filter contains the complement of every singleton. Thus N regards θ as measurable. $\square$

Corollary 8.5 (The high-HCD conclusion without an upper compact). Assume $\text{SC}(\delta)$, $\delta < \lambda \leq \gamma$, and $\text{CEX}_\gamma(\lambda)$. Let $\eta \geq q(\gamma)$ and assume explicitly that $N = \text{HCD}(\eta)$ satisfies ZFC. Then Theorem 8.1 and Corollary 8.2 hold for N , and every ambient regular $\theta > (2^\lambda)^N$ is measurable in N . In particular this holds for every ambient regular $\theta \geq (2^\lambda)^{+V}$.

Proof. Propositions 3.4 and 4.3 supply the hypotheses of Theorem 8.1. Apply Theorem 8.4. For the final bound, put $\chi = (2^\lambda)^N$. A bijection in N from χ to $(\mathcal{P}(\lambda))^N$ remains an injection in V into $\mathcal{P}^V(\lambda)$. Hence $|\chi|^V \leq 2^\lambda$ and $\chi < (2^\lambda)^{+V}$. This is an ordinal comparison; it does not assert that the two models calculate 2^λ identically. $\square$

Remark 8.6 (A second route from the quantitative theorem to no grounds). A set-forcing ground has arbitrarily high successor cardinals that remain regular in the extension. Choose one above $(2^\lambda)^N$. Theorem 8.4 would make it measurable in N , even though it is a successor there. This is another way to see the ground obstruction. The direct covering proof is shorter and does not need filter atomicity; the atomicity argument explains the stronger structure forced on any remaining high inner model.

9 A weaker lower compactness assumption

The contradiction only needs to cover a countable cofinal set. Consequently one may weaken the lower strongly compact hypothesis without changing the upper one.

Definition 9.1 (The countably complete fine-ultrafilter hypothesis). For an uncountable cardinal δ , let $F_{\omega_1}(\delta)$ mean: for every sufficiently large regular θ there is a countably complete fine ultrafilter on $P_\delta(\theta)$.

Proposition 9.2 (The countable version of the seed argument). Replace strong compactness in Theorem 5.2 by the existence of a countably complete fine ultrafilter on $P_\delta(\theta)$. Its conclusion then holds with “countably complete” in place of “ δ -complete,” and every countable $x \subseteq \mu$ has a cover in N of N -size below δ .

Proof. The well-founded ultrapower still has the seed u with $j''\theta \subseteq u$ and $|u|^M < j(\delta)$. The cofinality bounds, the transferred filter trace, the set $s \in N^j$, its M -smallness, and inclusion $j''\mu \subseteq s$ are proved exactly as in Theorem 5.2. They do not use $\text{crit}(j) \geq \delta$. Continuity at countable cofinality uses only $j(\omega) = \omega$, which holds for the well-founded ultrapower.

The derived ultrafilter is countably complete: given a countable sequence of its members, j fixes the index set ω , so s lies in the image of their intersection. For a countable x , only countably many fine cones must be intersected with the concentrating set. This produces the required cover; Lemma 2.1 again supplies the internal size bound. No claim of δ -completeness is made in this version. $\square$

Theorem 9.3 (Weakened no-sandwich theorem). There are no cardinals $\delta < \lambda \leq \gamma < \kappa$ such that $F_{\omega_1}(\delta)$, $CE_{\gamma}(\lambda)$, and $SC(\kappa)$ all hold. More generally, under $F_{\omega_1}(\delta)$ and $CE_{\gamma}(\lambda)$ with $\delta < \lambda$, no $HCD(\eta)$ for $\eta \geq q(\gamma)$ is a set-forcing ground of V .

Proof. If such a high model were a ground, choose θ in Lemma 6.1 large enough for $F_{\omega_1}(\delta)$ to apply. Proposition 9.2 covers a countable cofinal subset of λ by an internally $< \delta$ set in that model. The inherited regularity barrier contradicts this. An upper strongly compact κ then supplies the forbidden ground by Input C. $\square$

For a familiar sufficient hypothesis, let δ be regular and ω_1 -strongly compact, in the sense that every δ -complete filter extends to a countably complete ultrafilter. On $P_\delta(\theta)$, the filter generated by the sets

$$\{b \in P_\delta(\theta) : a \subseteq b\} \quad (a \in P_\delta(\theta))$$

is proper and δ -complete: regularity of δ ensures that the union of fewer than δ many such a 's still has size below δ . A countably complete ultrafilter extending it is fine. Thus this regular ω_1 -strong compactness assumption implies $F_{\omega_1}(\delta)$ and can replace the lower strong compactness in the main result. We do not assert an equivalence between these formulations at singular δ .

10 Scope, proof audit, and remaining questions

10.1 What has and has not been established

The main theorem is a conditional inconsistency of a specific *joint* large-cardinal configuration. It is not merely an inconsistency obtained by appending the Ground Axiom or an unproved stabilization principle: both were removed. The upper and lower strongly compact cardinals supply different, indispensable inputs to the proof. The upper one gives a high-completeness ground; the lower one supplies a fine ultrafilter used to derive countable covering in that ground.

The bare theory

$$ZFC + \exists \delta < \lambda \leq \gamma (SC(\delta) \wedge CE_{\gamma}(\lambda))$$

is not refuted by this report. Problem 5.2 in the Blue–Goldberg notes asks about precisely that remaining configuration [2]. Our conclusion is that any such model must have no strongly compact above its cover bound, no high-HCD ground, and—whenever a high HCD model satisfies Choice—the stationary-width and measurability behavior of Section 8.

Nor does the argument apply automatically to ordinary exactingness, to ultraexactingness without a cover hypothesis, or to a hypothesis whose cover bound changes with the height. The fixed bound γ is used in the trace reconstruction and is the reason an upper $\kappa > \gamma$ can be selected uniformly.

10.2 Dependency audit

Ingredient	Source or proof	Role
Completeness barrier	Inherited from the supplied synthesis; reconstructed in Section 3 from Inputs A–B.	Makes λ regular in the high model.
High HCD is a ground	Goldberg, Input C; requires the upper strongly compact.	Converts the inner model into a set-forcing ground.
ω -club amenability	Proposition 4.3, by hereditary definability.	Places the stationary-positivity trace in the high model.
Seed belongs to N^j	Equation (5.4); Separation from the trace.	Makes the derived measure concentrate on N .
Seed is small	Equations (5.3)–(5.5); disjointness gives an injection into a small club.	Provides $< \delta$ covers.
High stationary family	Lemma 6.1 and Appendix A.	Uses only that the forcing is a set.
No-sandwich contradiction	Theorems 7.1 and 1.1.	New application; no stabilization or Ground Axiom.
Quantitative measurability	Theorem 8.1, trace completeness, and Appendix B.	A stronger local consequence, not needed for the main contradiction.

10.3 Sensitive points checked explicitly

The inequality in trace reconstruction is $|\sigma| \leq \gamma < \eta$; it is not $|\sigma| < \gamma$. The ultrafilters used to define high-HCD sets have completeness above the cover bound, whereas the stationary-seed ultrafilter comes from the *lower* compact cardinal and has completeness δ . They are different ultrafilters with different jobs.

The model N^j is the interpretation in M of the definition of N . It is not $j''N$. The club-filter trace used there is calculated in M , rather than in N^j . Its membership in N^j follows by transferring the actual amenability statement. Merely saying that stationarity is definable would not justify the proof. The ordinal β need not be regular in M ; amenability was required at every ordinal of uncountable cofinality, not just at regular cardinals.

No ground stationarity is assumed to be preserved at a singularized ordinal or its first old successor. The proof moves above the size of the forcing. The Ulam-matrix and chain-condition proofs in Appendix A show exactly why that move is legitimate.

The upper compact cardinal is assumed strongly compact in V , not merely in a ground. The lower compact cardinal is likewise an ambient hypothesis. Choice in a high HCD model is obtained from its being a ground or is explicitly assumed; it is not inherited from the ambient Axiom of Choice alone.

10.4 Research questions left by the result

The central remaining question is whether the lower-compact, cover-exacting configuration can exist when the strongly compact cardinals are bounded. A more focused obstruction would be to derive that some $\text{HCD}(\eta)$ with $\eta \geq q(\gamma)$ is a ground without assuming an upper strongly compact. The present work proves that any such derivation would finish the inconsistency argument, but does not supply it.

Another precise target is the Choice question for the high-completeness models under the lower-compact hypothesis alone. The measurable-cardinal conclusion above is conditional on Choice there.

Eliminating that qualification would require an argument about the inner model, not a change of notation in the stationary-seed proof.

The supplied Lean sources were used as a reference for the hypotheses and the inherited barrier. No new Lean theorem is included, and no compilation of those sources is represented as a verification of the new result. The present proof obligations are conventional mathematical ones, isolated in the table above. The bibliographic search and source checks support the attribution of the inputs, not a claim that no one has previously noticed this consequence.

A Classical stationary facts used in the ground argument

A.1 Splitting at a successor by an Ulam matrix

Lemma A.1 (Successor splitting). In ZFC, if ν is infinite, $\theta = \nu^+$, and $S \subseteq \theta$ is stationary, then S has a family of θ pairwise disjoint stationary subsets. The family can be made a partition of S .

Proof. For every $\beta \in [\nu, \theta)$ choose a bijection $e_\beta : \nu \rightarrow \beta$. For $\xi < \theta$ and $i < \nu$, put

$$A_i^\xi = \{\beta < \theta : \beta > \max\{\nu, \xi\} \text{ and } e_\beta(i) = \xi\}.$$

For fixed ξ , the sets A_i^ξ as i varies partition a tail of θ . At least one $S \cap A_i^\xi$ is stationary: otherwise intersect the fewer than θ many clubs witnessing nonstationarity, contradicting stationarity of S . Choose one such $i = i(\xi)$.

There is a fixed $i_* < \nu$ for which $B = \{\xi < \theta : i(\xi) = i_*\}$ has size θ . If every fiber had size below the regular cardinal θ , their union over $\nu < \theta$ many indices would have size below θ , although that union is θ . For distinct $\xi, \zeta \in B$, the sets $A_{i_*}^\xi$ and $A_{i_*}^\zeta$ are disjoint, because $e_\beta(i_*)$ cannot have two values. Enumerating B gives the required stationary family. Any unused part of S can be added to one cell to obtain a partition, without affecting disjointness or stationarity. $\square$

The proof is applied *inside the ground*. At a regular uncountable θ there, E_ω^θ is stationary: given a club, build a strictly increasing countable sequence from it and take its supremum, which is still below θ and belongs to the club. The supremum has cofinality ω .

A.2 Chain-condition preservation, with names

Lemma A.2 (θ -cc preservation). Suppose $W \models \text{ZFC}$, θ is regular uncountable in W , and $\mathbb{P} \in W$ is θ -cc in W . Then forcing with $\mathbb{P}$ preserves regularity of θ and every stationary $S \subseteq \theta$ of W . It also preserves all W -cardinals at or above θ .

Proof. All choices of antichains in this proof are made in W . For a name $\dot{f} : \rho \rightarrow \theta$ with $\rho < \theta$, a maximal antichain deciding each $\dot{f}(i)$ has size below θ . Regularity of θ bounds the union of all possible values for all $i < \rho$ below θ . Thus no new such map can be cofinal, and θ stays regular.

Now let p force that $\dot{C}$ is club in θ . Work below p . For each $\alpha < \theta$, choose a maximal antichain B_α whose members decide an ordinal $\beta_q > \alpha$ belonging to $\dot{C}$. This is possible because $\dot{C}$ is forced unbounded. The chain condition gives

$$f(\alpha) = \sup\{\beta_q + 1 : q \in B_\alpha\} < \theta.$$

Let $D \subseteq \theta$ be the ground-model club of nonzero limit ordinals ξ closed under f , meaning $f(\alpha) < \xi$ for all $\alpha < \xi$. In any generic extension containing p , for each $\alpha < \xi \in D$ the generic meets B_α , so $\dot{C}$ has a point strictly between α and ξ . Closedness therefore gives $\xi \in \dot{C}$. Thus p forces $D \subseteq \dot{C}$. If S is stationary in W , choose $\xi \in S \cap D$ in W . Then p forces that S meets $\dot{C}$. Since the name and condition were arbitrary, S remains stationary.

For cardinal preservation, suppose $\chi \geq \theta$ is a W -cardinal and $\rho < \chi$. In a name for a function from ρ to χ , there are fewer than θ possible values per argument, by maximal antichains. If $\rho < \theta$, their union has size below θ ; if $\rho \geq \theta$, it has W -size at most $|\rho|^W \cdot \theta$ and hence below χ . In neither case can the function be onto χ . No cardinal at or above θ is collapsed. $\square$

B The filter-atomicity lemma, with proof

This is the classical Ulam lemma used in [4, Lemma 2.1]. It is included so that the quantitative measurability consequence does not rest on an unexplained atomicity step. The lemma is applied inside N , where Choice is available.

Lemma B.1 (Ulam atomicity). Let λ be an infinite cardinal and F a proper filter on a set X . Suppose F is $(2^\lambda)^+$ -complete and there is no family of λ pairwise disjoint F -positive sets. Then X can be partitioned modulo the dual ideal into fewer than λ positive atoms. Consequently F is an intersection of fewer than λ ultrafilters, each inheriting every degree of completeness of F .

Proof. Let $I = \{A \subseteq X : X \setminus A \in F\}$ be the dual ideal and put $\rho = (2^\lambda)^+$. A positive set A is an *atom* when for each $B \subseteq A$, either $B \in I$ or $A \setminus B \in I$. A positive non-atom can therefore be split into two disjoint positive sets.

Suppose no partition modulo I into fewer than λ positive atoms exists. We construct refining partitions modulo I through stages $\alpha \leq \lambda$. Start with $\{X\}$. At a successor stage, choose a non-atom cell and split it into two positive cells, leaving the others alone. There must be a non-atom cell by the supposition. Every partition we obtain has fewer than λ positive cells, by the assumed weak saturation.

At a limit stage $\alpha \leq \lambda$, take all intersections along coherent paths through the preceding partitions. There are at most $\lambda^\lambda = 2^\lambda$ such paths. Their intersections partition what has not already been discarded as ideal-small. Discard every intersection that is in I . The union of all discarded sets remains in I : there are at most $2^\lambda < \rho$ candidates at each stage and at most λ stages, and I is ρ -complete. Thus the positive intersections still form a partition modulo I , again with fewer than λ cells. They refine the preceding partitions by actual inclusion. This justifies the recursion, including its final stage λ .

Let $\mathcal{D}$ be that final partition; it has size $\tau < \lambda$. Every positive cell at every earlier stage contains a positive member of $\mathcal{D}$. Indeed $\mathcal{D}$ covers X modulo I and refines that stage; a positive cell with no final positive descendant would be contained in the discarded ideal-small set. For each of the λ successor-stage splits, choose one final cell inside each of its two children. This associates to the split an unordered pair of distinct members of $\mathcal{D}$. Pairs associated to different splits must differ: the two members of such a pair are in the same cell immediately before that split and in distinct cells afterward, so the split is the unique first stage at which they are separated. We have obtained an injection from λ into $[\mathcal{D}]^2$, impossible because $|[\mathcal{D}]^2| \leq \tau^2 < \lambda$. The recursion must therefore have reached a partition into positive atoms before this contradiction.

Let $\{A_i : i < \tau\}$, $\tau < \lambda$, be such a partition modulo I . For each i define

$$U_i = \{B \subseteq X : A_i \setminus B \in I\}.$$

The atom property gives the ultrafilter alternatives, and U_i is proper since $A_i \notin I$. If I is closed under unions of fewer than a cardinal ζ , then U_i is ζ -complete: intersecting fewer than ζ members has complement inside A_i contained in the corresponding union of I -small complements. Finally $F = \bigcap_{i < \tau} U_i$. One inclusion is immediate. For the other, if all $A_i \setminus B$ are in I , then

$$X \setminus B \subseteq \left(X \setminus \bigcup_{i < \tau} A_i \right) \cup \bigcup_{i < \tau} (A_i \setminus B) \in I,$$

using $\tau < \lambda < \rho$. Hence $B \in F$, as required. $\square$

C Source map and reproducibility

The starting archive contains the two reports docs/Large_Cardinals_Synthesis.tex and docs/Large_Cardinals_Unified_Report.tex, together with the Cardinals/Lean development. The specific inherited material used here is the synthesis's completeness barrier, its relative-predicate lemmas, and its high-HCD ground discussion. The new argument does not depend on the archive's more speculative Prikry consistency refinement or on its ultraexacting uniformization results.

Primary sources were checked on 18 September 2026. For the ground theorem, the exact version cited is arXiv:2103.13961v2. For the ω -club-amenable covering dichotomy, the checked author PDF is dated 10 August 2021 and has the result as Lemma 2.7. Its fuller singular-cardinal dichotomy is Corollary 2.9; only Lemma 2.7 is needed for our short cross-check. For cover exactingness, the checked notes are dated 1 July 2026. These version details matter because some statement numbers, and the strength of hypotheses used in short summaries, differ.

The accompanying source is self-contained as a $\text{\LaTeX}$ document: its bibliography is embedded and it needs no separate figure or bibliography files. Compile with `pdflatex` three times, or with `latexmk -pdf`. The text and mathematical fonts are the same `newpxtext/newpxmath` family as in the supplied synthesis, and the five named colors and page geometry are retained. No font files are distributed. The build checks concern rendering and cross-references, not the logical correctness of the mathematics.

References

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